Md Muntaqim Meherab

B.Sc. in Computer Science & Engineering

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LinkedIn GitHub Google Scholar

Profile

Undergraduate Computer Science & Engineering student and research assistant at DIU's NLP & ML Research Lab. My research focuses on remote sensing, climate change AI, and reliable machine learning, alongside deep learning for vision, LLM evaluation, and neuro-symbolic reasoning. I enjoy turning ideas into working implementations, running careful experiments, and writing results clearly so others can reproduce and build on them.

Education

B.Sc. in Computer Science & Engineering
Daffodil International University, Dhaka, Bangladesh

Publications & Research Outputs

- **Zero-Gradient Mid-layer Evolutionary Homophilic Exploration for Remote-sensing Adaptation with Frozen Backbones** 2027
  Meherab et al. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Main Conference. Project.*
- **Does Confidence Track Visibility? Diagnosing Blind Confidence in Cloud-Occluded Flood Segmentation** 2026
  Meherab et al. *NeurIPS 2026 Workshop: Tackling Climate Change with Machine Learning (TCCML). Project.*
- **What Your CXR Reveals Beyond the Pathology: Causal Evidence for Demographic Leakage in Medical Foundation Model Embeddings and Its Calibration Fairness Consequences** 2026
  Meherab et al. *ECCV 2026 Workshop: MEDFMB. Project.*
- **CoSSA: Correlation-Structure Shift Adapter for Cross-City Urban Forecasting** 2025
  Md Muntaqim Meherab *NeurIPS 2025 Workshop: UrbanAI (co-organised by Google). Paper.*
- **Gist Meets Graph: A Dual-Trace Neuro-Symbolic Reasoner for Knowledge Graph Question Answering** 2026 (to appear)
  Md Muntaqim Meherab *AAAI 2026 Workshop: NeusymBridge. Springer Nature (edited volume).*
- **Axiomatic Negation Coherence in Language Models: Evidence from FOLIO** 2026
  Meherab et al. *AAAI 2026 Bridge Program: LMReasoning.*
- **Advancing NLP Equity: A Secondary Benchmark Evaluation of Multilingual Language Models for Underrepresented Languages** 2026 (to appear)
  Meherab et al. *AAAI 2026 Workshop: LM4UC. Springer (LNCS/CCIS; indexed in Scopus, DBLP, EI Compendex). Paper.*
- **Multi-Domain Patch-Dropout: Improving Robustness Under Missing Data** 2025
  Meherab et al. *5th International Conference on Trends in Electronics and Health Informatics (TEHI 2025), Springer.*
- **Dynamic Region-Aware Gradient Suppression (DRAGS): Enhancing Vision Model Robustness by Suppressing Noisy Feature Regions During Training** 2025 (accepted)
  Meherab et al. *International Conference on Intelligent Data Analysis and Applications (IDAA 2025), Atlantis Press (Springer Nature). Paper.*

Books

- **Safe & Smart AI Prompts: Schema-First Prompt Engineering for LLMs**.
  Amazon KDP, 2025
- **Python Made Simple: For Everyone**.
  Amazon KDP, 2026

Awards & Achievements

- Library Excellence Award — Central Library, DIU (Best Library User; First Position).
- Champion Team — Data Structures course project; awarded by the Dean, FSIT, DIU.
- Champion Team — Object-Oriented Programming course project (*Smart Agro System*).

Leadership & Service

- **Academic Service:** Reviewer, ICLR 2026 Workshop on LLM Reasoning (OpenReview).
- **Academic Service:** Reviewer, ICLR 2026 2nd Workshop on World Models (OpenReview).
- **Academic Service:** Reviewer, ICLR 2026 Workshop Agents in the Wild: Safety, Security, and Beyond.
- **Volunteer Service:** Volunteer, 4th National Data Science Summit.
- Daffodil AI Club — Assistant General Secretary.
- DIU Data Science Club — Executive Member.
- DIU Reader's Club — Former General Secretary.

Technical Skills

Programming: Python, C/C++, Java
Frameworks & Libraries: PyTorch, NumPy, Pandas, Scikit-learn
AI/ML: Deep Learning, LLMs (prompting & evaluation), Computer Vision, NLP
Tools: Google Colab, Jupyter Notebook, Git, Linux
Research: Academic writing, literature review, experiment documentation, reproducibility, LaTeX